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Professional Profile

Senior System Engineer with 5+ years of experience in electric drive systems, power electronics, and industrial automation within safety-critical environments. Specialized in inverter-based systems, electric machines, system architecture, and requirements engineering. Strong expertise in system integration, validation, and troubleshooting under real operating conditions. Experienced in cross-functional collaboration and international project environments, with a structured approach to delivering reliable and high-performance systems aligned with functional safety and system-level requirements.

Professional Experience

May 2023 – Present **System Engineer – Electric Drive Systems / Power Electronics, Schaeffler AG**
(Former Vitesco Technology AG), Nuremberg, Germany

- Define and manage system requirements for high-voltage inverter and electric drive systems (up to 800V), ensuring traceability and alignment with customer and regulatory requirements.
- Develop system architectures and coordinate integration of electrical, electronic, and software components across domains.
- Apply motor control concepts (FOC, PWM) to analyze inverter-machine interaction and system performance.
- Perform system-level debugging, validation, and troubleshooting under nominal and fault conditions.
- Plan and support verification and validation activities using **MiL, SiL, HiL** and tools such as **MATLAB/Simulink** and **PLECS**.
- Support functional safety activities according to **ISO 26262**, including requirements derivation and risk analysis (FMEA/FTA).
- Contribute to EMC-related system design and validation activities.
- Collaborate with cross-functional teams (hardware, software, validation, safety) to ensure seamless integration.
- Facilitate technical discussions and design reviews within international teams.
- Work with system engineering tools such as **IBM DOORS** and **Siemens Polarion**.

October 2022 – April 2023 **Master Thesis – Power Electronics & Control Systems, Vitesco Technologies Germany GmbH**, Regensburg, Germany

- Analyzed inverter topologies (two-level VSI, T-type) and semiconductor technologies (Si, SiC, GaN) for electric drive applications.
- Implemented and validated motor control strategies (Space Vector PWM, FOC).
- Conducted multi-domain simulations linking electrical, thermal, and control behavior of inverter-machine systems.
- Evaluated system performance to support engineering decisions.

- February 2022 – **Intern – E-Mobility System Simulation**, *Vitesco Technologies Germany GmbH*, Regensburg, Germany
- September 2022
- Supported development of simulation models for electrified powertrain and electric drive systems.
 - Performed data analysis using **MATLAB** and **Python** to support system validation and performance analysis.
 - Assisted in integration of subsystems including converters, battery interfaces, and control strategies.
- July 2018 – **Automation Engineer – Drive Systems & Industrial Automation**, *Descon Engineering Limited*, Lahore, Pakistan
- September 2020
- Designed and commissioned variable frequency drive (VFD) systems for industrial applications (pumps, fans, conveyors).
 - Configured motor control modes (V/f, vector control) and protection features for reliable drive operation.
 - Integrated drive systems with **Siemens S7-1200/1500 PLCs** and SCADA systems using PROFINET and Modbus communication.
 - Developed HMI/SCADA interfaces for monitoring system performance, alarms, and operational parameters.
 - Performed commissioning, testing, and troubleshooting of drive systems at customer sites.
 - Provided technical support and optimized system performance under real operating conditions.

Education

- November 2020 – April 2023 **M.Sc. Electrical Engineering and Information Technology**, *Hochschule Darmstadt*, Darmstadt, Germany
- Focus Areas: Power Electronics, Electrical Drives, Embedded Systems, Control Engineering
 - Thesis: Multi-domain modelling and validation of inverter systems
 - CGPA: 1.8 (German Grade System)
- September 2014 – April 2018 **B.E Electrical Engineering**, *Hamdard University*, Karachi, Pakistan
- Focus Areas: Power Electronics, Electrical Drives, Power Systems
 - CGPA: 1.3 (German Grade System)

Technical Skills

- Systems & Drives Electric Drive Systems, Inverters, Motor Control (FOC, PWM), System Integration, Validation (MiL/SiL/HiL)
- Simulation MATLAB/Simulink, PLECS, PSCAD, ANSYS Twin Builder
- Automation PLC (TIA Portal, S7-1200/1500), SCADA/HMI, PROFINET, Modbus, Ethernet
- Engineering Requirements Engineering, Functional Safety (ISO 26262), Interfaces, DOORS, Polarion
- Programming Python, MATLAB, Basic C/C++

Language

- English:** Professional working proficiency (C1)
- German:** Intermediate working proficiency (B2)